AADK Task 2 – Compose Essentials
Build, Analyze, Refine

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Task: AADK Task 2 – Compose Essentials

Aim

To design and implement a basic Jetpack Compose UI screen using core layout primitives, apply responsive modifiers, analyze structural hierarchy, and refine the interface using accessibility and consistency principles.

Project Overview

In this task, a simple Compose-based UI screen was developed to demonstrate the practical use of foundational layout elements. The implementation showcases:

- Column (vertical layout structure)
- Row (horizontal grouping of elements)
- Box (conceptual understanding of layered layouts)
- Spacer (controlled spacing)
- Modifier (layout behavior control)
- Padding and Alignment
- Responsive layout techniques

The UI was structured with emphasis on clarity, visual hierarchy, spacing consistency, and adaptability across screen sizes.

Implementation Code

```
@Composable
fun ProfileScreen() {
Column(        modifier
= Modifier
          .fillMaxSize()
          .padding(16.dp),
      horizontalAlignment = Alignment.CenterHorizontally
   ) {

      Text(
```

```
        text = "Welcome to Compose",
fontSize = 22.sp,
        fontWeight = FontWeight.Bold
    )

    Spacer(modifier = Modifier.height(16.dp))

    Row(
        modifier = Modifier.fillMaxWidth(),
        horizontalArrangement = Arrangement.SpaceEvenly
    ) {
        Button(onClick = { }) {
            Text("Like")
        }

        Button(onClick = { }) {
            Text("Share")
        }
    }
  }
}
```

Layout Primitives Used

1. Column
   Arranges UI elements vertically and serves as the main container of the screen.
2. Row
   Places elements horizontally and is used here to align the action buttons side by side.
3. Box (Conceptual Usage)
   Used for stacking elements on top of each other. Suitable for overlay UI such as badges or floating elements.
4. Spacer
   Introduces controlled spacing between elements and improves readability.

Responsive Modifiers Applied

The following modifiers were used to ensure responsiveness and layout stability:

- fillMaxSize() – Expands layout to occupy the entire screen.
- fillMaxWidth() – Makes Row span across screen width.
- padding(16.dp) – Adds consistent outer spacing.
- height(16.dp) – Controls vertical spacing using Spacer.
- Alignment.CenterHorizontally – Centers content horizontally.
- Arrangement.SpaceEvenly – Distributes buttons evenly across available width.

These modifiers help maintain consistency across different screen sizes.

UI Analysis

The layout follows essential UI design principles:

Visual Hierarchy
The heading appears first and uses bold styling with larger font size, clearly indicating importance.

Balanced Spacing
Uniform 16dp spacing ensures the layout is clean and uncluttered.

Logical Grouping
The buttons are grouped inside a Row to indicate related actions.

Alignment Consistency
Center alignment provides symmetry and enhances readability.

Overall, the layout demonstrates structured organization and intuitive interaction flow.

Accessibility Considerations

The UI incorporates fundamental accessibility principles:

- Adequate heading text size for readability.
- Clear and meaningful button labels.
- Sufficient spacing between interactive elements.
- Avoidance of crowded layouts.

Recommended Improvements:

- Ensure minimum 48dp touch target size for buttons.
- Add contentDescription for screen reader compatibility if icons are introduced.
- Verify color contrast according to accessibility standards.

Refinement Suggestions

To further improve the design:

- Integrate Material3 theme for consistent styling.
- Introduce a defined color palette using MaterialTheme.

- Apply a structured spacing system (for example, 8dp increments).
- Add consistent corner radius for buttons.
- Test layout across multiple device sizes, including tablets.
- Use weight modifier where proportional distribution is needed.

Conclusion

This task improved understanding of Jetpack Compose layout primitives, responsive modifiers, and structured UI development. It reinforced concepts of visual hierarchy, spacing consistency, and accessibility awareness. The exercise strengthened both technical implementation skills and design refinement thinking, forming a strong foundation for advanced Compose development.